

CYBERNETICS

MADE EASY:

1.The Basic Control Loop

BY

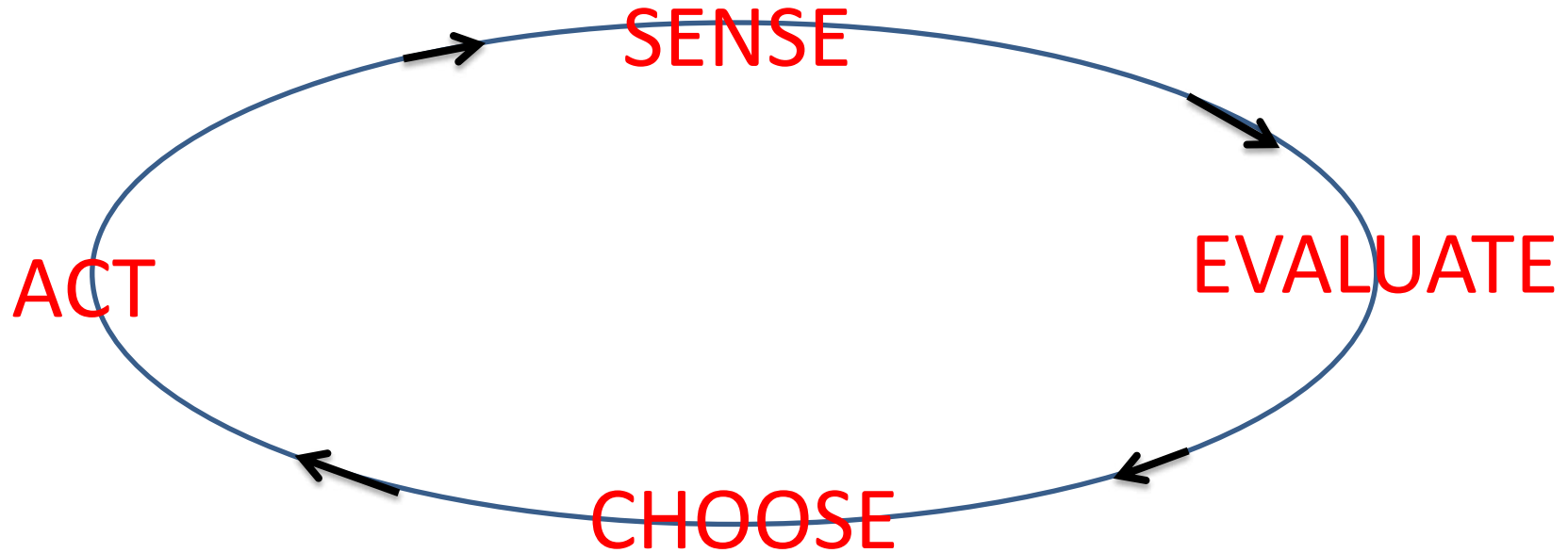
ROBERT JOHANNSON

The term “cybernetics” comes from the Greek word “KYBERNAMA” which means to steer, to guide or to govern. In plain English it is the theory of control.

We are all familiar with the basic control process.

- 1.You do something. (ACT)
- 2.You see how it turned out. (SENSE)
- 3.You compare the results to what you wanted. (EVALUATING)
- 4.You choose another action to bring it closer to your goal. (CHOOSE)
- 5.You do something else. (ACT)

The Cybernetic Control Loop



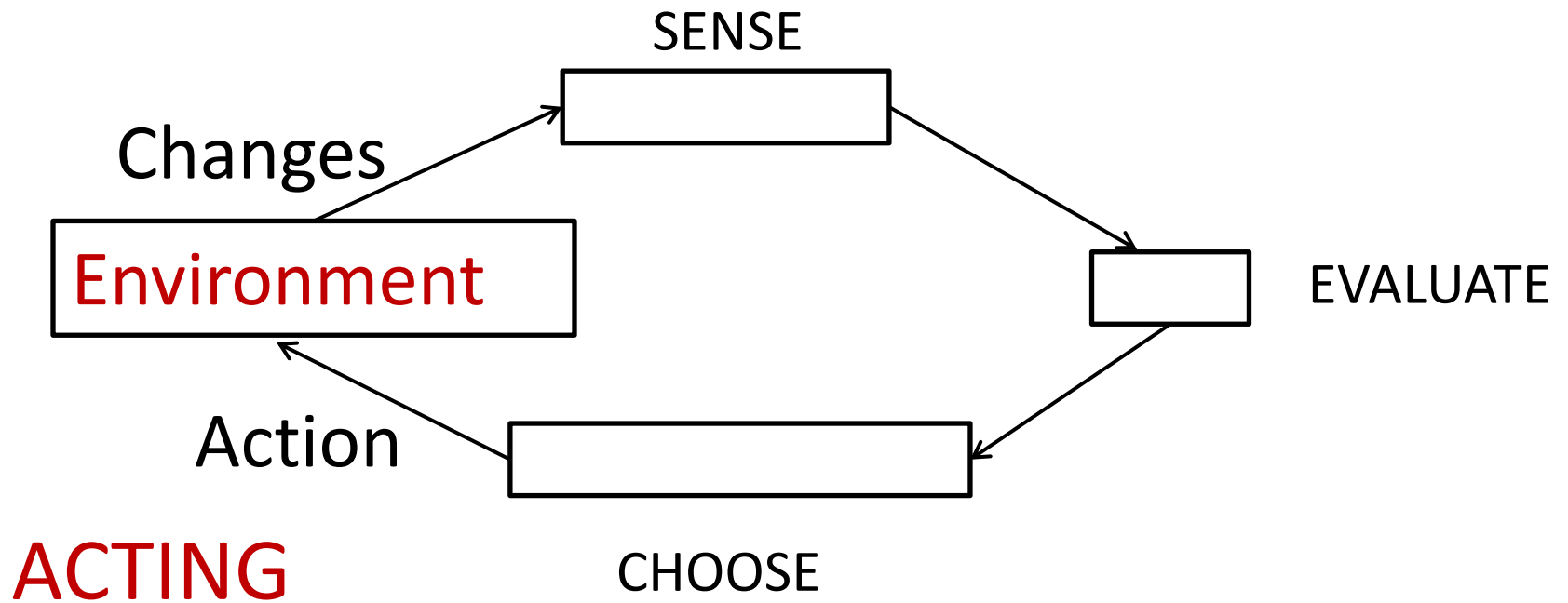
These four processes form a loop. Any control process consists of going around this basic control loop.

Each of the individual processes around the control loop is an information process.

An information process consists of translating one form of information into another form of information.

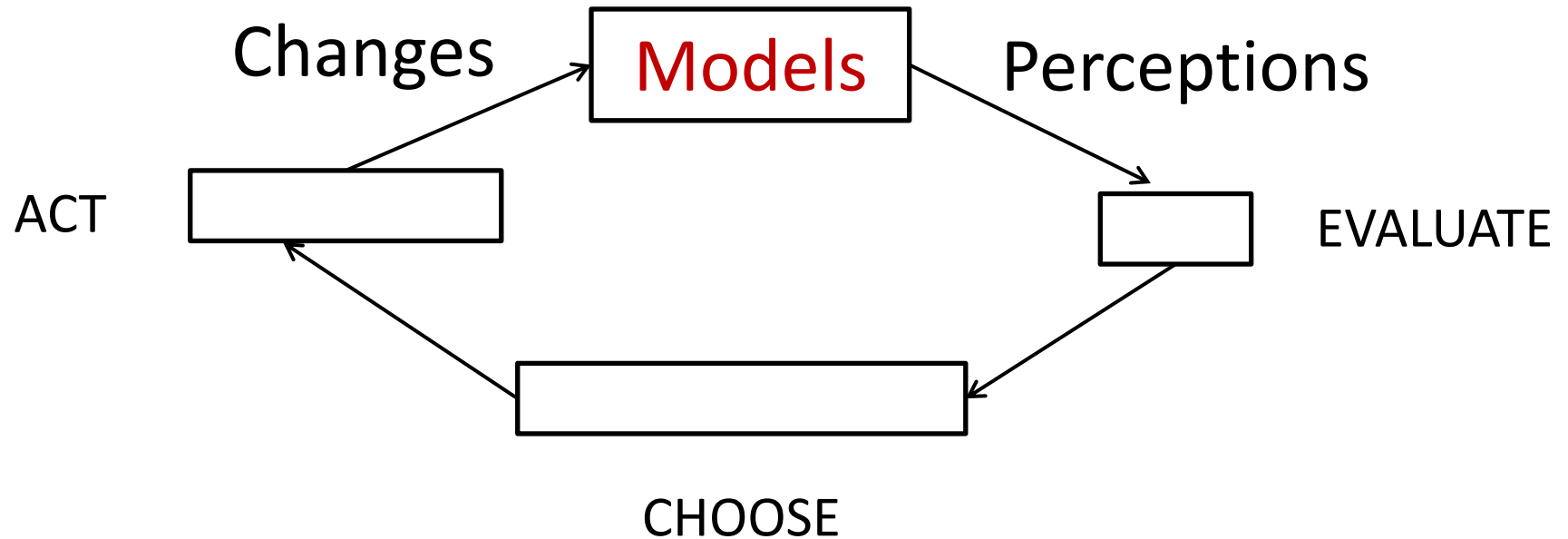
We can use the box as a symbol of the translation process, and the arrow as a symbol of the information.



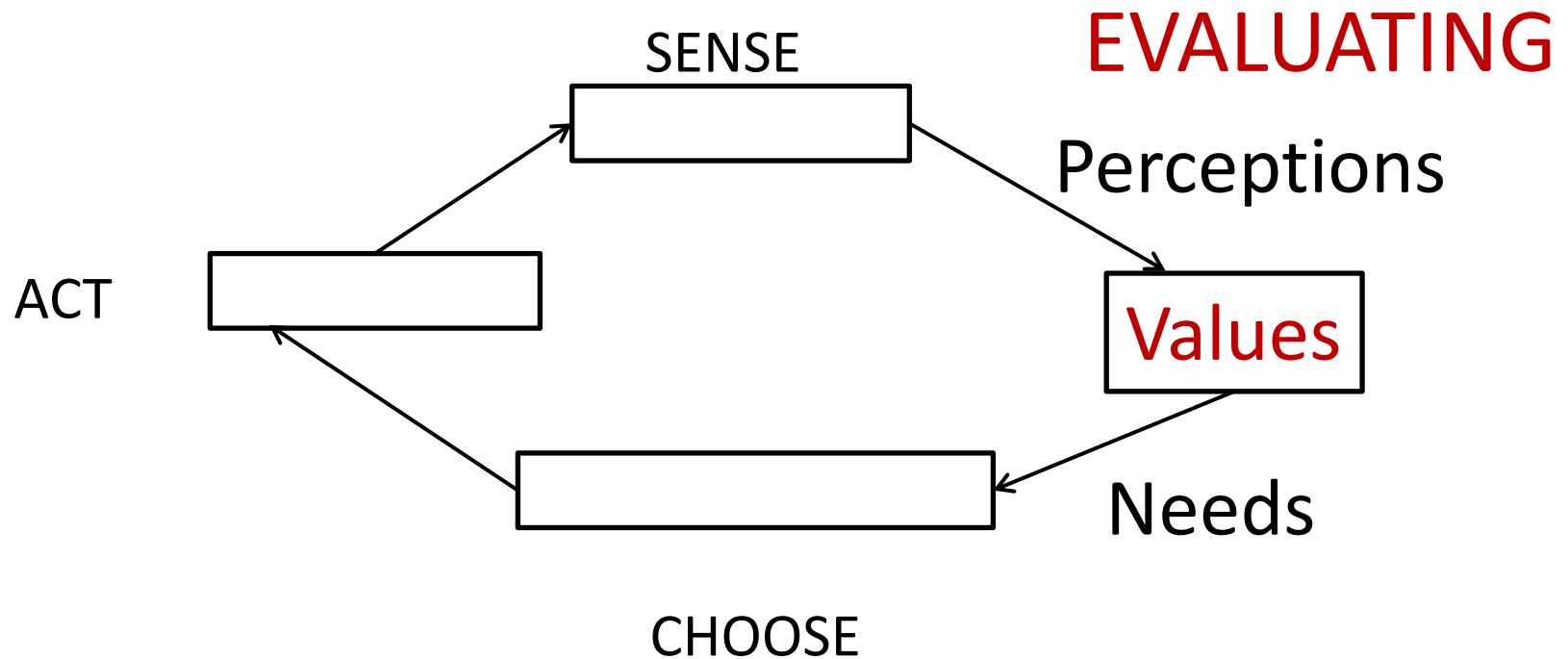


Acting consists of our action being translated by the **environment** into results or changes.

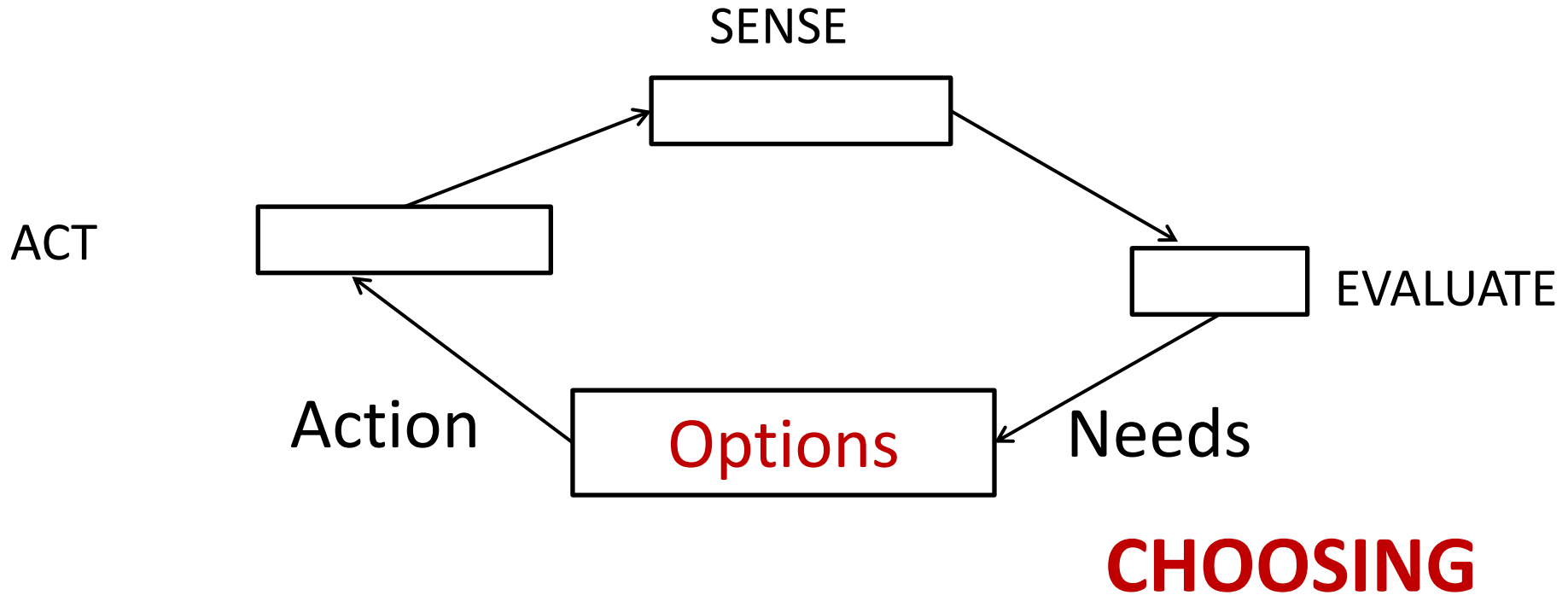
SENSING



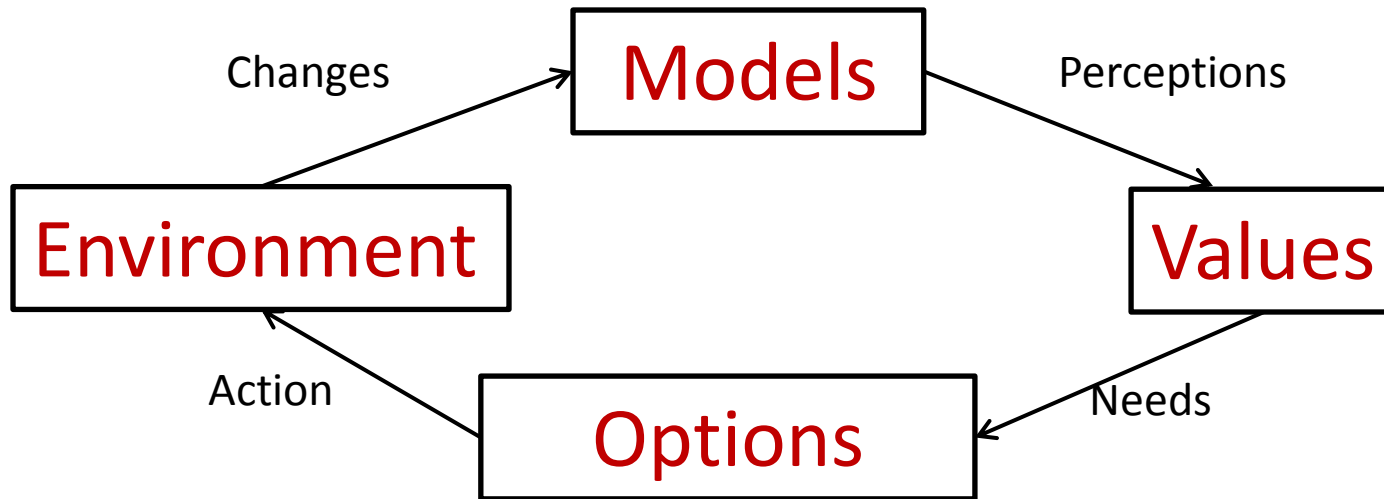
Sensing consists of the changes being translating by our **models** into perceptions.



Evaluating consists of our perceptions being translated by our **values** into needs.



Choosing consists of our needs being translated by our **options** into action.



The translation processes are information. They are commands about how to process information, in other words programs. The translation processes constitute the program that controls the control process.

This is a general model of control. It applies to all control processes. To analyze any control system or decision making system there are four questions:

1. What part of the environment is the system trying to control?
2. What is the model they are using to understand the environment?
3. What values are they attempting to maintain?
4. What are the options available to them?